Orthopedics Power System

（JDXY-DGD-01）

INSTRUCTION MANUAL

Tianjin Jindong Xiyi Medical Instrument Factory

License Key：津食药监械生产许 20100054 号

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Preface

Dear Customer:

Tianjin Jindong Xiyi Medical Instrument Factory is a professional manufacturer of electric surgical instruments. In particular, our neurosurgical and orthopedic instruments are always at the leading level compared with similar products in China. The modernized factory is located in the Tianjin New Technology Industrial Park, where scientific research institutes gather and talents are concentrated. Its own strong R&D and production capabilities enable it to continue producing medical instruments with excellent quality and advanced technology to serve clinical subjects.

First we would like to express our appreciation for you choosing our products. We hope that our products will help you in your work. In order to make you skilled in operating this system, please read this manual carefully before use.

Note:

The information in this manual is subject to change without prior notice. Please understand.

1. Overview

The power system adopts the power supply mode. The whole machine can be sterilized with high temperature and high pressure, and the power is strong. It does not generate heat for long run and works with stepless speed regulation, built-in brushless motor, high speed, low noise and stable performance.
The power system is designed for microcomputer control and modularization. The connectors can be operated without errors and are safe and easy.
The product is suitable for neurosurgery craniotomy. This power system combines drilling, milling and grinding function in one machine. Based on a security device the craniotomy drill stop automatically after drilling through. The new milling cutter is made of special materials, which makes up the defects of traditional milling cutters and really improves the cutting speed.

2. Technical Parameters

1	Charger input voltage	220V 50Hz
2	Output power	200W
3	Torque >	1600g. cm
4	Hardness	HRC58
5	Drill working speed	0～1500 rpm
6	Milling cutter working speed	0～60000 rpm
7	Surgery drill working speed	0～60000 rpm
8	Noise is less than	75dba

Safety performance indicators comply with the *"GB9706.1 Medical Electrical Equipment Part I, General Safety Requirements"* .

3. Operating Environment

Meets the requirements of the national standard GB/T14710

- Environmental temperature 5°C～40°C
- Relative humidity ≤80%

4. Use of Host Controller

(1). Switch on the power. There is a power input interface on the back of the controller, use the power cable to connect 220V power. There are two connection jacks in the lower right corner of the front of the controller. The first one is for the foot switch and the second one is for the motor handle.

The two sets of indicators are for mode selection (left) and steering selection (right). M1 is the standard starting state (as the default starting mode), M2 grinding drill starting state (in this state the handle starts smoothly); the steering default is F which is the positive rotation. R is the reverse rotation state (Note: Crab drills, milling cutters must not run in this state).

The right-hand knob *Speed Limit Setting* is the speed setting key, which can limit the required maximum speed. (Note: The knob rotates to the left and its speed adjustment range is 0 - 40% of the maximum speed of the machine. The knob rotates to the maximum position and its speed adjustment range is 0 - the maximum speed of the machine)

(2). After the power cable, foot pedal and motor handle are inserted, the front left lower corner switch can be turned on to the "I" position, each indicator light is on, and the speed display is "0". The working state at this moment is mode 1, forward rotation. Rotate the knob right to set the required maximum rotation speed. If you need to reverse, press F/R key, R indicator light is on; if you need mode 2, press M1/M2 key, M2 indicator light is on. The foot switch is stepless. The heavier the food swith is pressed, the higher is the speed. To restore the initial state, press the *Reset* key and turn the *Speed Limit Setting* button left to zero.

(3). The power output is a smart jack. The jack can be connected to different sizes of motors, and the corresponding operating parameters can be automatically edited for different sizes of the inserted motor.

5. Instruction

(1). The motor handle, cutting tools, drills, tools etc. should be sterilized under high temperature and high pressure before use.

Note: Before the motor handle is subjected to high temperature and pressure sterilization, the disinfection protection cover must be screwed and sealed on the front end of the motor handle to prevent water vapor and dirt from entering the motor and causing damage! Do not use the motor handle continuously. If you find that the handle is overheated, stop using it immediately. After the motor handle is locked, immediately release the foot switch or turn the speed setting knob to “0”. Do not frequently start the motor handle to avoid overheating.

(2). Insert the cable plug of the foot switch into the corresponding socket on the host (Each socket of the machine only matches specific part of the system). and insert the cable plug of the moter handle into the corresponding socket on the host. Connect the power cable with the host and power supply.

(3). Turn the switch to the "I" position and the power indicator lights up.

(4). When selecting a craniotome drill, first connect the booster with the hand-held motor handle, and then connect the craniotomy drill with the booster. When installing a craniotomy drill, pull the puller to connect with the output shaft of the booster and insert it. Turn the craniotomy drill so that the opening aligns to the position and release the pull sleeve. Meantime, check that the craniotomy drill should not be pulled out.
(5). Take the booster as an example, first make the booster unlocked. Then insert the booster into the interface of the hand-held motor handle, so that the protruding portion of the interface and the recess of the hand-held motor handle interface are inserted into each other. After that turn the booster sleeve to make it locked; when disassembling, unlock and unplug it. The same method applies for installing other working heads.
(6). Handling of Milling Cutters: The milling cutter can be loaded and unloaded in a quick and easy way without using tools. First unscrew the craniotomy outer locking sleeve and then remove the milling cutter holder. Hold the milling cutter in one hand and use the thumb and forefinger to pull the skull milling shaft downward. Meanwhile, use the other hand to pull out and remove the milling cutter needle. Insert the changed milling cutter needle and release the thumb and forefinger. Then install the cutter holder and lock the locking sleeve.
Note: During the surgery, the apron must be installed on the milling cutter needle to avoid the blood flowing into the interior of the skull, which may cause corrosion .
(7). Handling of the sharpening tool: The drill cutter can be loaded and unloaded in a quick and easy way without using tools. Hold the drill handle in the left hand, use the thumb and index finger to hold the pull sleeve and gently push it up. Then insert the grinding head with right hand and release the pull sleeve. Remove the grinding head the other way around.
(8). Hold the handle stably during operation, use force evenly and speed up gradually. Familiarize yourself with the performance and simulate operation before using it.

6. Cautions

(1). This set of equipment has a filter and it must be well grounded during use.
(2). Before use, check whether all parts are normal, whether there is any abnormal sound or abnormal vibration, and whether the rotating part is smooth.
(3). Before the more handle is subjected to high temperature and high pressure sterilization, the disinfection protection cover must be screwed and sealed on the front end of the motor handle to prevent water vapor and dirt from entering the motor interior and causing damage.
(4). Before surgery, check whether the craniotomy drill can stop automatically.
(5). Before the operation, check whether the working head is sharp enough and whether the installation is firm and proper.
(6). Before the operation, check whether the main unit, motor handle and working parts are operating normally, and whether the accessories and consumables are complete. If the motor handle jams or abnormal operation occurs during operation, the power should be turned off immediately, clear trouble and re-test. The system can be put back into use only when it works normally again.
(7). The system must be operated by trained physicians to prevent injury.

Note: The whole machine can not be soaked.

7. Maintenace

(1). After use, the blood should be cleaned and wiped, oiled and maintained in time to avoid damage.
(2). The machine should be stored in a cool, dry and ventilated place.
(3). The machine must avoid strong vibrations, collisions, and rough handling during transportation.
(4). When troubles occur, promptly contact the manufacturer or get the machine repaired by maintenance personnel who are authorized by the manufacturer.

8. Packing List

See the list.

9. How to clean cutting tools

Before use, the instrument head must be properly installed, then get it wrapped with a protective cloth and sterilized under high temperature and pressure (134°C, 2 atm, 3.5 min); The motor handle should be formalined or applied with a one-time sterile drill bushing. Or get the tools protected with a sterile protective cloth cover after sterilization under high temperature and pressure (134°C, 2 atmospheres, 3.5 min).

Warranty

Dear Customer,

First of all, thank you for choosing products. In order to make our service more satisfactory to you, please read this manual carefully after purchase and keep well the product warranty card attached to this manual.

Warranty instructions:

1. Please fill in the information enclosed with the warranty card on this manual and complete the seal of the final direct distributor. If the seal is not available, please contact the original seller to get the seal to cover your rights. Please be sure to keep the invoice or copy. Otherwise, we will refer to the date of manufacture as a reference for warranty.
2. Our products have one-year guarantee and life-long maintenance. If there is a problem within one year after the product is sold, we will repair the product for free. (Buys bear the freight). If there is a problem due to human factors or out of the warranty period, we will repair it and may charge a reasonable amount of maintenance cost.
3.When problems occur, please contact our after-sales service department or your product distributor immediately. We will respond within 24 hours and do not disassemble the product by yourself.

Contact：

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